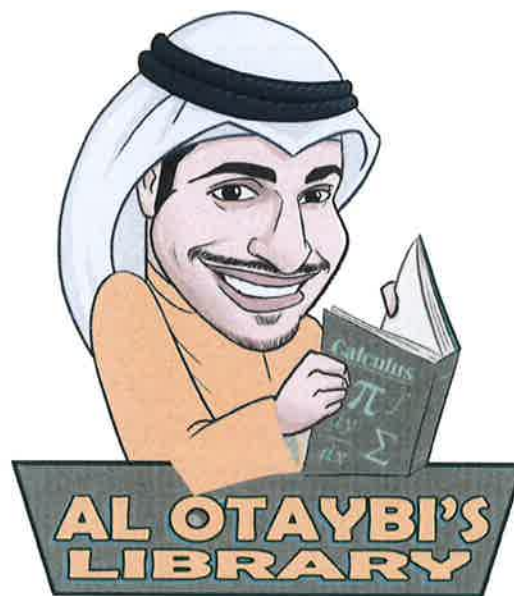


Kuwait University

2021/2022

Spring course



Calculus B

Second

Section : 11.1 (Part c/d)

11.1 Sequences.

(8 c)



11.1 Sequences



Limit of sequences given by recurrence relation

if $\{a_n\}$ is convergent

Then

$$\lim_{n \rightarrow \infty} a_n = L$$

$$\lim_{n \rightarrow \infty} a_{n+1} = L$$

$$\lim_{n \rightarrow \infty} a_{n-1} = L$$



Example

Let a sequence given by the recurrence relation

$$2a_{n+1} = 5 + \frac{3}{n} + a_n, \quad a_1 = 2$$

Assuming that a_n is convergent, Find $\lim_{n \rightarrow \infty} a_n$

$\therefore a_n$ is convergent $\begin{cases} \therefore \lim_{n \rightarrow \infty} a_n = L \\ \therefore \lim_{n \rightarrow \infty} a_{n+1} = L \end{cases}$

$$\lim_{n \rightarrow \infty} 2a_{n+1} = \lim_{n \rightarrow \infty} \left[5 + \frac{3}{n} + a_n \right]$$

$$2L = 5 + \frac{3}{\infty} + L$$

$$2L - L = 5 + 0$$

$$\boxed{L = 5}$$

Find the limit of the sequence $\{a_n\}$ with

$$a_1 = 2 \quad \text{and} \quad 3(a_{n-1})^2 a_n = 2(a_{n-1})^3 - 8 \quad \forall n \geq 2$$

Given that a_n is convergent

$$\because a_n \text{ is convergent} \begin{cases} \therefore \lim_{n \rightarrow \infty} a_n = L \\ \therefore \lim_{n \rightarrow \infty} a_{n-1} = L \end{cases}$$

$$\lim_{n \rightarrow \infty} 3(a_{n-1})^2 a_n = \lim_{n \rightarrow \infty} [2(a_{n-1})^3 - 8]$$

$$3L^2 L = 2L^3 - 8$$

$$3L^3 = 2L^3 - 8$$

←

$$3L^3 - 2L^3 = -8$$

$$L^3 = -8$$

$$\Rightarrow \therefore L = -2$$

Suppose that $\{a_n\}$ is a sequence with $a_n > 0$ for all n and

$$a_n = \frac{8}{3} + \frac{1}{a_{n-1}}$$

if $\lim_{n \rightarrow \infty} a_n = L$, Find L .

$$\therefore \lim_{n \rightarrow \infty} a_n = L \longrightarrow \therefore \lim_{n \rightarrow \infty} a_{n-1} = L$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left[\frac{8}{3} + \frac{1}{a_{n-1}} \right]$$

$$L = \frac{8}{3} + \frac{1}{L}$$

$$L = \frac{8L}{3L} + \frac{3}{3L}$$

$$L = \frac{8L + 3}{3L}$$

$$3L^2 = 8L + 3$$

$$3L^2 - 8L - 3 = 0$$

$$(3L + 1)(L - 3) = 0$$

$$L = -\frac{1}{3}$$

rejected

$$\boxed{L = 3}$$

A convergent sequence is given by

$$a_1 = \frac{1}{2} \text{ and } a_{n+1} = \frac{1}{2} \left(a_n + \frac{1}{a_n} \right), \text{ for } n \geq 1.$$

a) List the first three terms of $\{a_n\}_{n=1}^{\infty}$.

b) Find the limit of $\{a_n\}_{n=1}^{\infty}$. (L70)

Dec 2015

a) $a_1 = \frac{1}{2}$

$$\begin{aligned} n=1 &\longrightarrow a_2 = \frac{1}{2} \left(a_1 + \frac{1}{a_1} \right) = \frac{1}{2} \left(\frac{1}{2} + \frac{1}{\frac{1}{2}} \right) \\ &= \frac{1}{2} \left(\frac{1}{2} + 2 \right) = \frac{1}{2} \left(\frac{1}{2} + \frac{4}{2} \right) = \frac{1}{2} \cdot \frac{5}{2} = \frac{5}{4} \end{aligned}$$

$$\begin{aligned} n=2 &\longrightarrow a_3 = \frac{1}{2} \left(a_2 + \frac{1}{a_2} \right) = \frac{1}{2} \left(\frac{5}{4} + \frac{1}{\frac{5}{4}} \right) \\ &= \frac{1}{2} \left(\frac{5}{4} + \frac{4}{5} \right) = \frac{1}{2} \left(\frac{25}{20} + \frac{16}{20} \right) = \frac{1}{2} \cdot \frac{41}{20} = \frac{41}{40} \end{aligned}$$

b) $\because a_n$ is convergent $\begin{cases} \therefore \lim_{n \rightarrow \infty} a_n = L \\ \therefore \lim_{n \rightarrow \infty} a_{n+1} = L \end{cases}$

$$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} \frac{1}{2} \left(a_n + \frac{1}{a_n} \right)$$

$$L = \frac{1}{2} \left(L + \frac{1}{L} \right)$$

$$L = \frac{1}{2} \left(\frac{L^2 + 1}{L} \right)$$

$\times 2L$

$$2L^2 = L^2 + 1$$

$$L^2 = 1$$

$$\therefore L = 1$$

$L = -1$ rejected

Let $\{a_n\}$ be convergent sequence satisfying the relation

$$a_1 = \sqrt{2} \quad a_n = \sqrt{2 + a_{n-1}} \quad n \geq 2$$

(a) Find the first four terms of the sequence.

(b) Find $\lim_{n \rightarrow \infty} a_n$

(a) $a_1 = \sqrt{2}$

$n=2 \rightarrow a_2 = \sqrt{2+a_1} = \sqrt{2+\sqrt{2}}$

$n=3 \rightarrow a_3 = \sqrt{2+a_2} = \sqrt{2+\sqrt{2+\sqrt{2}}}$

$n=4 \rightarrow a_4 = \sqrt{2+a_3} = \sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2}}}}$

(b) $\because a_n$ is convergent $\begin{cases} \therefore \lim_{n \rightarrow \infty} a_n = L \\ \therefore \lim_{n \rightarrow \infty} a_{n-1} = L \end{cases}$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sqrt{2 + a_{n-1}}$$

$$L = \sqrt{2 + L}$$

$$L^2 = 2 + L$$

$$L^2 - L - 2 = 0$$

$$(L + 1)(L - 2) = 0$$

$$L = -1$$

rejected

$$L = 2$$



Tricky question !!



Let $\{a_n\}$ be sequence defined by

$$a_0 = a_1 = 1 \quad \text{and} \quad a_{n+1} = a_n + a_{n-1} \quad \forall n$$

Given that $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = L$ exist.

Find the value of L

$$\therefore \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = L \longrightarrow \therefore \lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}} = L$$

$$a_{n+1} = a_n + a_{n-1}$$

$$\div a_n \quad \frac{a_{n+1}}{a_n} = 1 + \frac{a_{n-1}}{a_n}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \left[1 + \frac{a_{n-1}}{a_n} \right]$$

$$L = 1 + \frac{1}{L}$$

$$L = \frac{L}{L} + \frac{1}{L}$$

$$L = \frac{L+1}{L}$$

$$L^2 = L+1$$

$$L^2 - L - 1 = 0$$

$$a=1 \quad b=-1 \quad c=-1$$

$$L = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{1 \pm \sqrt{1 - 4(1)(-1)}}{2(1)} = \frac{1 \pm \sqrt{5}}{2}$$

$$\Rightarrow \therefore L = \frac{1 + \sqrt{5}}{2}$$



Multiple choice



If a convergent sequence is defined $a_1 = 1$ and

Fall 2018

$$a_{n+1} = \frac{1}{2 + a_n}, \quad n \in \mathbb{N}, \quad \text{then } \lim_{n \rightarrow +\infty} a_n =$$

a) $\sqrt{2} + 1$ b) $\sqrt{2} - 1$ c) 1 d) -1 e) None of the above.

$\therefore a_n$ is convergent $\begin{cases} \therefore \lim_{n \rightarrow \infty} a_n = L \\ \therefore \lim_{n \rightarrow \infty} a_{n+1} = L \end{cases}$

$$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} \frac{1}{2 + a_n}$$

$$L = \frac{1}{2 + L}$$

$$L(2 + L) = 1$$

$$2L + L^2 = 1$$

$$L^2 + 2L - 1 = 0$$

$$a = 1 \quad b = 2 \quad c = -1$$

$$L = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-2 \pm \sqrt{4 - 4(1)(-1)}}{2(1)}$$

$$= \frac{-2 \pm \sqrt{8}}{2} = \frac{-2 \pm \sqrt{4(2)}}{2}$$

$$= \frac{-2 \pm 2\sqrt{2}}{2} = \cancel{2} \frac{[-1 \pm \sqrt{2}]}{\cancel{2}}$$

$\Rightarrow \therefore L = -1 + \sqrt{2}$ (b) ✓



Monotonic



Monotonic

لا ثبات أن السيكونيس

decreasing

أو

increasing

يتم اثبات انها

عن طريق

إذا كانت صعبة
الاشتقاق

الاشتقاق

$$a_{n+1} - a_n > 0$$

increasing

or

$$a_{n+1} - a_n < 0$$

decreasing

$$f'(x) > 0$$

increasing

or

$$f'(x) < 0$$

decreasing



Bounded



Bounded

لا ثبات أن السيكونيس

عن طريق اللمت

Convergent

يتم اثبات انها

If a_n is convergent then $\Rightarrow a_n$ is bounded

العكس غير صحيح

$$m \leq a_n \leq M$$

Lower bound
bounded below

Lower bound
bounded below

Determine whether the sequence is monotonic or not. where

$$a_n = \frac{n}{n^4 + 1}, \quad n \geq 1$$

Let $f(x) = \frac{x}{x^4 + 1}$

$$f'(x) = \frac{(x^4 + 1)(1) - x \cdot (4x^3)}{(x^4 + 1)^2}$$

$$= \frac{x^4 + 1 - 4x^4}{(x^4 + 1)^2}$$

$$= \frac{1 - 3x^4}{(x^4 + 1)^2} < 0 \quad \forall x \in [1, \infty)$$

$\Rightarrow \therefore f(x)$ is decreasing on $[1, \infty)$

$\Rightarrow \therefore a_n$ is decreasing $\Rightarrow \therefore a_n$ is monotonic.

Determine whether the sequence is bounded or not. where

$$a_n = \frac{n}{n+1}, \quad n \geq 1$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n}{n+1} = \lim_{n \rightarrow \infty} \frac{\cancel{n}}{\cancel{n}(1 + \frac{1}{n})} = \frac{1}{1+0} = 1$$

$\Rightarrow \therefore a_n$ is convergent

$\Rightarrow \therefore a_n$ is bounded.

$$\frac{1}{2} \leq a_n < 1$$

Lower bound
(bounded below)

Upper bound
(bounded above)

Determine whether the sequence is monotonic or not ?

$$a_n = \frac{n}{n+1}$$

Let $f(x) = \frac{x}{x+1}$

$$f'(x) = \frac{(x+1)(1) - x \cdot (1)}{(x+1)^2}$$

$$= \frac{x+1 - x}{(x+1)^2} = \frac{1}{(x+1)^2} > 0 \quad \forall x \in [1, \infty)$$

$\Rightarrow \therefore f(x)$ is increasing

$\Rightarrow \therefore a_n$ is monotonic.

Determine whether the sequence is monotonic or not ?

$$a_n = e^n$$

Let $f(x) = e^x$

$$f'(x) = e^x > 0 \quad \forall x \in [1, \infty)$$

$\therefore a_n$ is increasing

$\Rightarrow \therefore a_n$ is monotonic.



Text book questions



Determine whether the sequence is increasing, decreasing, or not monotonic. Is the sequence bounded?

$$a_n = \frac{1}{2n+3}$$

Let $f(x) = \frac{1}{2x+3}$

$$f'(x) = \frac{(2x+3) \cdot 0 - (1) \cdot 2}{(2x+3)^2} = \frac{-2}{(2x+3)^2} < 0 \quad \forall x \in [1, \infty)$$

$\therefore f(x)$ is decreasing

$\therefore a_n$ is decreasing $\longrightarrow \therefore a_n$ is monotonic.

$\lim_{n \rightarrow \infty} \frac{1}{2n+3} = \frac{1}{\infty} = 0 \longrightarrow \therefore a_n$ is bounded $0 < a_n < \frac{1}{5}$

Determine whether the sequence is increasing, decreasing, or not monotonic. Is the sequence bounded?

$$a_n = \frac{1-n}{2+n}$$

Let $f(x) = \frac{1-x}{2+x}$

$$f'(x) = \frac{(2+x)(-1) - (1-x)(1)}{(2+x)^2}$$

$$= \frac{-2 - x - 1 + x}{(2+x)^2} = \frac{-3}{(2+x)^2} < 0 \quad \forall x \in [1, \infty)$$

$\therefore f(x)$ is decreasing

$\therefore a_n$ is decreasing $\longrightarrow \therefore a_n$ is monotonic.

$\lim_{n \rightarrow \infty} \frac{1-n}{2+n} = \lim_{n \rightarrow \infty} \frac{n[\frac{1}{n}-1]}{n[\frac{2}{n}+1]} = \frac{0-1}{0+1} = -1 \longrightarrow \therefore a_n$ is bounded $-1 < a_n < 0$



Text book questions



Determine whether the sequence is increasing, decreasing or not monotonic. Is the sequence bounded?

$$a_n = ne^{-n}$$

Let $f(x) = xe^{-x}$

$$\begin{aligned} f'(x) &= (1)e^{-x} - xe^{-x} \\ &= \frac{1}{e^x} - \frac{x}{e^x} \end{aligned}$$

$$= \frac{1-x}{e^x} < 0 \quad \forall x \in (1, \infty)$$

$\Rightarrow \therefore f(x)$ is decreasing

$\Rightarrow \therefore a_n$ is decreasing $\Rightarrow \therefore a_n$ is monotonic.

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} ne^{-n}$$

$$\lim_{n \rightarrow \infty} \frac{n}{e^n} \quad \text{g/b}$$

Let $f(x) = \frac{x}{e^x}$

$$\lim_{x \rightarrow \infty} \frac{x}{e^x}$$

L'HOP

$$\lim_{x \rightarrow \infty} \frac{1}{e^x} = \frac{1}{\infty} = 0$$

$\Rightarrow \therefore a_n$ is convergent $\Rightarrow \therefore a_n$ is bounded

$$0 < a_n \leq \frac{1}{e}$$



Text book questions



Determine whether the sequence is increasing, decreasing or not monotonic. Is the sequence bounded?

$$a_n = n + \frac{1}{n}$$

Let $f(x) = x + \frac{1}{x}$, $x \geq 1$

$$\begin{aligned} f'(x) &= 1 - \frac{1}{x^2} \\ &= \frac{x^2}{x^2} - \frac{1}{x^2} \end{aligned}$$

$$= \frac{x^2 - 1}{x^2} > 0 \quad \forall x \in (1, \infty)$$

$\Rightarrow \therefore f(x)$ is increasing

$\Rightarrow \therefore a_n$ is increasing $\Rightarrow \therefore a_n$ is monotonic

$$\lim_{n \rightarrow \infty} a_n$$

$$\lim_{n \rightarrow \infty} \left(n + \frac{1}{n} \right) = \infty + \frac{1}{\infty} = \infty + 0 = \infty$$

$\Rightarrow \therefore a_n$ is Divergent

$\Rightarrow \therefore a_n$ is unbounded.



Tricky question !!



Show that the sequence $a_n = \left\{ \frac{2^n}{n!} \right\}_{n=1}^{\infty}$ is monotonic and bounded.

$$a_n = \frac{2^n}{n!} \longrightarrow a_{n+1} = \frac{2^{n+1}}{(n+1)!}$$

$$\begin{aligned} a_{n+1} - a_n &= \frac{2^{n+1}}{(n+1)!} - \frac{2^n}{n!} \\ &= \frac{2^n \cdot 2}{(n+1) n!} - \frac{2^n}{n!} \\ &= \frac{2^n \cdot 2}{(n+1) n!} - \frac{2^n (n+1)}{n! (n+1)} \end{aligned}$$

$$= \frac{2^n \cdot 2 - 2^n (n+1)}{(n+1) n!}$$

$$= \frac{2^n [2 - (n+1)]}{(n+1) n!}$$

$$= \frac{2^n [2 - n - 1]}{(n+1) n!}$$

$$= \frac{2^n [1 - n]}{(n+1) n!} < 0 \quad \forall n \in (1, \infty)$$

$\Rightarrow \therefore a_n$ is decreasing $\Rightarrow \therefore a_n$ is monotonic.

$$\lim_{n \rightarrow \infty} \frac{2^n}{n!}$$

$$\frac{2^n}{n!} = \frac{(2) (2) (2) \dots (2) \cancel{(2)} (2)}{n (n-1) (n-2) \dots (3) \cancel{(2)} (1)}$$

$$0 < \frac{2^n}{n!} < \frac{2}{n} \quad (2) \rightarrow \text{check}$$

$$\lim_{n \rightarrow \infty} 0 = 0$$

$$\lim_{n \rightarrow \infty} \frac{2}{n} = \frac{2}{\infty} = 0$$

∴ By Sandwich Thm

$$\lim_{n \rightarrow \infty} \frac{2^n}{n!} = 0$$

⇒ ∴ a_n is convergent

⇒ ∴ a_n is bounded

$$0 < a_n \leq 2$$



Tricky question !!



Determine whether the sequence is increasing , decreasing or not monotonic . Is the sequence bounded ?

$$a_n = (-1)^n \frac{1}{n}$$

$$n=1 \longrightarrow a_1 = (-1)^1 \frac{1}{1} = -1$$

$$n=2 \longrightarrow a_2 = (-1)^2 \frac{1}{2} = (1) \frac{1}{2} = \frac{1}{2}$$

$$n=3 \longrightarrow a_3 = (-1)^3 \frac{1}{3} = -\frac{1}{3}$$

$$n=4 \longrightarrow a_4 = (-1)^4 \frac{1}{4} = \frac{1}{4}$$

$$\implies a_n = \left[-1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, \dots \right]$$

$$\therefore a_1 < a_2 > a_3 < a_4 \dots$$

$\implies \therefore a_n$ is not monotonic.

$$\lim_{n \rightarrow \infty} (-1)^n \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \left| (-1)^n \frac{1}{n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{1}{n} \right| = \frac{1}{\infty} = 0$$

$$\therefore \lim_{n \rightarrow \infty} (-1)^n \frac{1}{n} = 0$$

$\implies \therefore a_n$ is convergent $\implies \therefore a_n$ is bounded



Tricky question !!



Determine whether the sequence is increasing ,
decreasing or not monotonic . Is the sequence bounded ?

$$a_n = \sin n$$

Let $f(x) = \sin x$

$$f'(x) = \cos x$$

$$\because -1 \leq \cos x \leq 1$$

$\Rightarrow \therefore f(x)$ is not monotonic.

~~$\lim_{n \rightarrow \infty} \sin n$~~

$$-1 \leq \sin n \leq 1$$

$\Rightarrow \therefore a_n$ is bounded.



Multiple choice



Which of the following is true for the sequence

$$\{\cos(0), \cos(1), \cos(2), \dots\}$$

(Assuming that the pattern of the first few terms continuous)

- a) Increasing. b) Decreasing. c) Convergent.
d) Bounded. e) None of the above.

Dec 2017

$$a_n = \{ \underset{\textcircled{1}}{\cos(0)}, \underset{\textcircled{2}}{\cos(1)}, \underset{\textcircled{3}}{\cos(2)}, \dots \}$$

$$\therefore a_n = \cos(n-1)$$

$$-1 \leq \cos(n-1) \leq 1$$

$\therefore a_n$ is bounded (d) ✓



Multiple choice



The sequence $\left\{ \frac{1}{\sqrt[5]{n^2+1}} \right\}$ is :

Dec 2016

- a) increasing and divergent. b) increasing and convergent.
c) decreasing and divergent. d) decreasing and convergent.
e) alternating and divergent.

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt[5]{n^2+1}} = \frac{1}{\infty} = 0$$

$\Rightarrow \therefore a_n$ is convergent.

$$f(x) = \frac{1}{\sqrt[5]{x^2+1}} = (x^2+1)^{-\frac{1}{5}}$$

$$f'(x) = \frac{-1}{5} (x^2+1)^{-\frac{6}{5}} 2x$$

$$= \frac{-2x}{5 (x^2+1)^{\frac{6}{5}}} < 0 \quad \forall x \in [1, \infty)$$

$\Rightarrow \therefore a_n$ is decreasing

(d)
✓



Multiple choice



The sequence $\{\tan^{-1}n\}$ is

Dec 2017

- a) decreasing and its limit is 0. b) increasing and its limit is $\pi/2$.
c) decreasing and its limit is $-\pi/2$. d) divergent.
e) not bounded.

$$\lim_{n \rightarrow \infty} \tan^{-1}n = \tan^{-1}\infty = \frac{\pi}{2}$$

$$f(x) = \tan^{-1}x$$

$$f'(x) = \frac{1}{x^2 + 1} > 0 \quad \forall x \in [1, \infty)$$

$\Rightarrow \therefore a_n$ is increasing (b) ✓



Answer each of the following with true or false.
Justify your answer



If a sequence $\{a_n\}$ is convergent then

$$\lim_{n \rightarrow \infty} [a_{n+1} - a_n] = 0$$

$\because a_n$ is convergent $\begin{cases} \lim_{n \rightarrow \infty} a_n = L \\ \lim_{n \rightarrow \infty} a_{n+1} = L \end{cases}$

$$\Rightarrow \lim_{n \rightarrow \infty} [a_{n+1} - a_n] = L - L = 0$$

$\Rightarrow$ True

if $\lim_{n \rightarrow \infty} [a_{n+1} - a_n] = 0$

then the sequence $\{a_n\}$ is convergent

False

Let $a_n = \ln n \rightarrow a_{n+1} = \ln(n+1)$

$$\lim_{n \rightarrow \infty} [a_{n+1} - a_n] = \lim_{n \rightarrow \infty} [\ln(n+1) - \ln(n)]$$

$$\lim_{n \rightarrow \infty} \ln\left(\frac{n+1}{n}\right) = \lim_{n \rightarrow \infty} \ln\left(\frac{n(1+\frac{1}{n})}{n}\right) = \ln(1+0) = 0$$

but $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \ln n = \infty$

$\Rightarrow a_n$ is Divergent.



Answer each of the following with true or false.
Justify your answer



If $0 < a_n < 1$

then the sequence $\{a_n\}$ is convergent

False

Let $a_n = \begin{cases} \frac{1}{3} & \text{if } n \text{ is even} \\ \frac{1}{2} & \text{if } n \text{ is odd} \end{cases}$

$\lim_{n \rightarrow \infty} a_n = \begin{cases} \frac{1}{3} & \text{if } n \text{ is even} \\ \frac{1}{2} & \text{if } n \text{ is odd} \end{cases}$

$\Rightarrow \therefore \lim_{n \rightarrow \infty} a_n \text{ DNE} \Rightarrow \therefore a_n \text{ is Divergent}$

if $\lim_{n \rightarrow \infty} a_n = 1$

then $\lim_{n \rightarrow \infty} (a_n)^n = 1$

False

Let $a_n = 1 + \frac{1}{n}$

$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right) = 1 + 0 = 1$

$\lim_{n \rightarrow \infty} (a_n)^n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e \neq 1$